
O-RAN Work Group 11 (Security work group)

Study on Security Configuration Management

Copyright © 2025 by the O-RAN ALLIANCE e.V.

The copying or incorporation into any other work of part or all of the material available in this document in any form without the prior written permission of O-RAN ALLIANCE e.V. is prohibited, save that you may print or download extracts of the material of this document for your personal use, or copy the material of this document for the purpose of sending to individual third parties for their information provided that you acknowledge O-RAN ALLIANCE as the source of the material and that you inform the third party that these conditions apply to them and that they must comply with them.

O-RAN ALLIANCE e.V., Buschkauler Weg 27, 53347 Alfter, Germany
Register of Associations, Bonn VR 11238, VAT ID DE321720189

Contents

List of figures	3
List of tables	3
Foreword.....	4
Modal verbs terminology	4
1 Scope	5
2 References	5
2.1 Normative references	5
2.2 Informative references	5
3 Definition of terms, symbols and abbreviations	6
3.1 Terms	6
3.2 Symbols	6
3.3 Abbreviations	6
4 SCM Background.....	6
4.1 Introduction.....	6
4.2 Considerations	7
5 Study and identify the security gaps related to SCM.....	7
5.1 Assets	7
5.1.1 O-RAN Architecture Elements.....	7
5.1.2 O-RAN interfaces	7
5.2 Security Controls	8
5.3 Input recommendations.....	8
5.4 Threat template	8
5.5 SCM Threats analysis	8
5.5.1 Certificate related threats	8
5.5.2 NACM related threats	9
5.5.3 LDAP related threats	9
6 Study and recommendation related to SCM	9
7 Conclusion.....	9
Annex: Change history/Change request (history)	10

List of figures

No table of figures entries found.

List of tables

No table of figures entries found.

Foreword

This Technical Report (TR) has been produced by WG11 of the O-RAN ALLIANCE.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN Alliance modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

version xx.yy.zz

where:

- xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.
- yy: the second digit-group is incremented when editorial only changes have been incorporated in the document. Always 2 digits with leading zero if needed.
- zz: the third digit-group included only in working versions of the document indicating incremental changes during the editing process. External versions never include the third digit-group. Always 2 digits with leading zero if needed.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

1 Scope

The main objective for this work item is to identify and develop the security management YANG models for O-RAN framework fulfilling the security requirements specified in [i.2]. The specification will be based on the YANG data modeling language (see RFC 7950). Security YANG models in O-RAN will apply to relevant O-RAN interfaces (O1, Open Fronthaul M-plane). This work item will perform the following steps,

- Identify if there is any existing security management YANG model available in O-RAN specifications
- Study and recommend the security requirement for YANG model aligning with [i.2]
- Study and identify the gaps related to security configuration
- Study the YANG models for certificate configuration
- Study the YANG models for NETCONF access control such as NACM configuration
- Study the YANG models for LDAP configuration

2 References

2.1 Normative references

Not applicable.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

- [i.1] O-RAN WG11: "O-RAN Security Protocols Specification".
- [i.2] O-RAN WG11: "O-RAN Security Requirements and Controls Specification".
- [i.3] O-RAN WG11: "O-RAN Threat Modelling and Risk Assessment".
- [i.4] O-RAN WG11: "O-RAN Security Test Specifications".
- [i.5] O-RAN WG4: "O-RAN Management Plane Specification".
- [i.6] IETF RFC 7950: "The YANG 1.1 Data Modeling Language".
- [i.7] O-RAN WG10: "O-RAN Information Model and Data Models Specification".
- [i.8] O-RAN WG10: "O1 Network Resource Model Specification".
- [i.9] O-RAN WG4: "Cooperative Transport Interface Transport Management Procedures Specification".
- [i.10] O-RAN WG5: "O1 Interface specification for O-DU".
- [i.11] O-RAN WG5: "O1 Interface specification for O-CU-UP and O-CU-CP".
- [i.12] An extensible YANG validator and converter in python, <https://github.com/mbj4668/pyang>.
- [i.13] O-RAN WG1: "O-RAN Architecture Description".
- [i.14] O-RAN WG10: "O-RAN Operations and Maintenance Interface Specification".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms [given in [i.1] and the following] apply:

Open Fronthaul M-Plane: Management interface controlling the O-RU, generally driven from the O-DU but in the case of the hybrid topology also driven from the SMO as specified in [i.13].

NOTE 1: Please refer to **Error! Reference source not found.** for more details.

O1: Interface used by the SMO framework, as specified in Clause 5.3.1, for operation and management of O-RAN NFs as specified in [i.13] [i.14].

O-Cloud: Cloud platform that provides O-RAN standardized interfaces, hosting O-RAN defined software components.

SMO: A Service Management and Orchestration framework.

NOTE 2: Please refer to Clause **Error! Reference source not found.** for more details as specified in [i.13].

3.2 Symbols

Void

3.3 Abbreviations

For the purposes of the present document, the abbreviations [given in [i.1] and the following] apply:

LDAP	Lightweight directory access protocol
NACM	NETCONF Access Control Model
NETCONF	Network Configuration Protocol
NF	Network Function
O-RAN	Open Radio Access Network
O-Cloud	O-RAN Cloud
PNF	Physical Network Function
SMO	Service Management and Orchestration
VNF	Virtualised Network Function
YANG	Yet Another Next Generation

4 SCM Background

4.1 Introduction

Management of security configuration is currently not handled in a consistent way in O-RAN. Different work groups see the topic in different ways. In some cases, this is expected to be a topic only for the security work group while in other cases it is defined within a work group.

For example, there are YANG models related to NACM included in WG4 definitions for Open Fronthaul M-plane. For O1, there are security requirements on NACM stated in [i.2]. There may be other areas where security configuration such as LDAP configuration, NACM configuration, Certificate Management configuration is required. Some of the security requirements related to NACM, LDAP are defined in [i.2].

There are several discussions on O1 security configuration aspects such as NACM, etc happening across O-RAN WGs (WG11, W10, etc) showing a need to have Security Configuration Management specifications in O-RAN.

4.2 Considerations

This document makes the following considerations:

- This TR performs a detailed study of the Security Configuration Management for O1 interface and Open Fronthaul M-plane.
- The existing O-RAN specifications such as O-RAN.WG10.Information Model and Data Models [i.7], O-RAN.WG10.O1NRM.1 [i.8], O-RAN.WG4.MP-YANGs [i.5], O-RAN.WG4.CTI-TMP-YANG [i.9], O-RAN.WG5.O-DU-O1.1 [i.10], O-RAN.WG5.O-CU-O1.0 [i.11] discuss the YANG models which are considered as the baseline specifications for the Security Configuration Management TR to identify and develop the security YANG models for O-RAN framework fulfilling the security requirements specified in [i.2].
- The Security YANG model considers the rules and guideline as specified in O-RAN specifications [i.5] [i.7] [i.8] [i.9] [i.10] [i.11].
- The pyang YANG validator tool is considered for evaluating the security YANG model as recommended in [i.12].
- The recommended security YANG models for O1 interface provided in this report to be shared with the impacted O-RAN Alliance working group, such as WG4 and WG10.

5 Study and identify the security gaps related to SCM

Editor's note: This clause will study and identify the security gaps related to security configuration management like NACM, certificate and LDAP aligning with SRCS and SPS.

5.1 Assets

This document makes the following considerations:

5.1.1 O-RAN Architecture Elements

- ASSET-C-01: Logical module: Service Management and Orchestration (SMO)
- ASSET-C-02: Near-RT RIC software
- ASSET-C-03: O-CU-CP software
- ASSET-C-04: O-CU-UP software
- ASSET-C-05: O-DU software
- ASSET-C-06: O-RU software
- ASSET-C-07: O-eNB
- ASSET-C-31: Shared O-RU

5.1.2 O-RAN interfaces

- ASSET-C-22: O1 interface, including protocol stack
- ASSET-C-24: Open FH M-plane, including protocol stack

5.2 Security Controls

The following security controls should be considered for the security gap analysis:

- Control-1: Certificate Management
- Control-2: TLS
- Control-3: mTLS
- Control-4: SSH
- Control-5: NACM with NETCONF
- Control-6: MACsec
- Control-7: IPsec
- Control-8: 802.1X

5.3 Input recommendations

The following are considered for security configuration management gap analysis:

- Certificate configuration
- NETCONF access control, NACM configuration
- LDAP configuration

5.4 Threat template

Threat ID	O-RAN threat ID
Threat title	[Threat Title will include suitable reference of 3GPP/IETF/ETSI]
Threat description	Introduction about the threat
Threat type	Spoofing Tampering Repudiation Information disclosure Denial of Service Elevation of Privilege
Vulnerability	TBD
Impact type	Authenticity Integrity Non-repudiation Confidentiality Availability Authorization
Affected Assets	Asset identification

5.5 SCM Threats analysis

5.5.1 Certificate related threats

Threat ID	T-SCM-Cert-01
Threat title	Configuration of incorrect Certificate parameter
Threat description	Certificate parameters may be misconfigured intentionally or unintentionally which may lead to the CA issuing a rogue certificate or the wrong certificate
Threat type	Denial of Service, Tampering
Vulnerability	It may lead to incorrect certificate issued by the CA issuer

Impact type	Availability
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-03, ASSET-C-04, ASSET-C-05, ASSET-C-06, ASSET-C-07, ASSET-C-31

Threat ID	T-SCM-Cert-02
Threat title	Tampering of Certificate parameter
Threat description	If the certificate parameters are tampered or comprised by the malicious actor, which may lead to initiation of valid certificate request to CA by the attacker
Threat type	Denial of Service, Tampering
Vulnerability	Attacker may claim as trusted entity
Impact type	Confidentiality Authenticity Availability
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-03, ASSET-C-04, ASSET-C-05, ASSET-C-06, ASSET-C-07, ASSET-C-31

5.5.2 NACM related threats

Void

5.5.3 LDAP related threats

Void

6 Study and recommendation related to SCM

Editor's note: This clause will study and recommend the security requirements and security usecases related to security configuration management like NACM, Certificate and LDAP aligning with SRCS and SPS.

Void

7 Conclusion

Editor's note : This clause will provide the summary for the SCM TR.

Void

Annex:

Change history/Change request (history)

Date	Revision	Description
2025.01.28	00.00.01	Skeleton proposal
2025.03.11	00.02	Approved CR in the month of feb and March 2025- Internal O-RAN release
2025.03.12	00.03	Feedback received during the SCM WI TR meeting [4] on 11/03/2025
2025.04.16	00.04	Approved CR in the month of April 2025 during the SCM WI TR meeting
2025.06.03	00.05	Approved CR in the month of June 2025 during the SCM WI TR meeting
2025.06.18	00.06	Updated the TR based on comments received and CR approved till June 2025 ERI-2025.02.11-WG11-CR0207-SCM-Clause1_Scope-v02.docx ERI-2025.02.11-WG11-CR0208-SCM-Clause2_Reference-v02.docx ERI-2025.02.24-WG11-CR0213-SCM-Clause3.1_Terms-v03.docx ERI-2025.02.24-WG11-CR0214-SCM-Clause3.3_Abbreviations-v01.docx ERI-2025.02.25-WG11-CR0211-SCM-Clause4_Background-v03.docx ERI-2025.02.25-WG11-CR0212-SCM-Clause4.1_Consideration-v03.docx ERI-2025.04.09-WG11-CR0226-SCM-Clause5.1_assets-v03.docx ERI-2025.04.09-WG11-CR0227-SCM-Clause5.2_SecurityControls-v04.docx ERI-2025.04.09-WG11-CR0228-SCM-Clause5.3_Recommendation for Security Configuration Management-v03.docx ERI-2025.05.07-WG11-CR0238-SCM-Clause5.4_Threats Template-v01.docx ERI-2025.05.07-WG11-CR0239-SCM-Clause5.5_SCMTThreatAnalysis-v01.docx ERI-2025.05.21-WG11-CR0244-SCM-Clause5.5.1-CertificateRelatedThreats-v02.docx